Regularization and Model Stability

Lecture 3

A predictor is a model input; its coefficient is a fitted weight.

Regularization adds a cost for large fitted weights.

Stability asks how sensitive the fit is to the training sample.

ECON 417 Business Forecasting

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Three Questions for a Fitted Model

- **Accuracy**: are forecasts close to later outcomes?
- **Stability**: how sensitive is the fit to the training sample?
- **Simplicity**: how many inputs and relationships must we explain?

Key message: One score does not answer all three questions.

Regression: Combining Inputs With Fitted Weights

$$\hat{y}_i = a + \beta_1 x_{i1} + \beta_2 x_{i2}.$$

- i: one observation; x_{i1}, x_{i2}: two inputs.
- a: intercept, or starting level.
- β_1, β_2: fitted weights, holding the other input fixed.
- $\hat{y}_i$: fitted prediction; y_i: observed outcome.

OLS Chooses the Best Training Fit

Ordinary least squares: OLS

Choose the intercept and weights to minimize training squared errors.

$$r_i = y_i - \hat{y}_i, \qquad \text{training loss} = \frac{1}{n}\sum_{i=1}^{n} r_i^2.$$

- A **residual** is an observed outcome minus its fitted value.
- n is the number of training observations.
- This is training fit, not evidence of future accuracy.

Two Kinds of Stability

Definitions

Coefficient stability: fitted weights change little across nearby samples.

Prediction stability: predictions change little for the inputs being examined.

Key message: Unstable individual weights do not automatically mean unstable predictions.

Same Prediction, Different Weights

$$\hat{y} = 10 + b_1 x_1 + b_2 x_2.$$

Model	b_1	b_2	Prediction at $(5, 5)$
A	2	0	20
B	0	2	20

Key message: Compare the combined prediction as well as each weight.

Multicollinearity: Inputs Share Information

Definition

One predictor can be closely approximated by a weighted sum of other predictors.

- Correlation measures straight-line movement between two variables.
  Strong pairwise correlation is one possible case.
- Individual contributions become difficult to separate.
- This need not prevent accurate predictions in familiar cases.

Why a Small Sample Change Can Matter

- Similar inputs allow several nearly equivalent fits.
- Replace a few observations: the preferred allocation may change.
- Examine predictions on the same reference inputs.
- Genuine changes in the relationship are a separate possibility.

Key message: A changing coefficient is a diagnostic, not a complete diagnosis.

Many Inputs Relative to the Sample

High-dimensional candidate set

Many candidate predictors relative to the observations available for fitting.

- A **feature** is another name for an input or predictor.
- More candidate inputs create more ways to fit accidental patterns.
- Availability at the forecast origin still applies to every input.

Bias and Variance Refer to Repeated Fits

- **Bias**: average fitted prediction differs from the underlying relationship.
- **Variance**: fitted predictions vary across hypothetical training samples.
- Restriction can reduce variance while increasing bias.
- Too much restriction can underfit.

Regularization Adds a Coefficient Cost

Definitions

Loss: score for prediction or fitting errors.
Penalty: added cost for a large or complex fitted solution.
Shrinkage: pulling fitted weights toward zero.

$$\text{fitting score} = \text{training loss} + \lambda\,\text{penalty}.$$

λ: penalty strength, chosen through validation.

A **hyperparameter** is chosen outside the coefficient-fitting step.

Ridge Uses Squared Coefficient Sizes

Ridge regression

Fit a regression while charging for the total squared slope weights.

$$\text{ridge score} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} \beta_j^2.$$

- n: training observations; p: predictor count.
- β_j: weight on predictor j; intercept excluded from penalty.
- The squared-size rule is called L_2.

Read the Score as a Trade-off

Candidate fit	A	B
Mean training squared error	4	5
Weights	$(2, 0)$	$(1, 1)$
Squared-weight total	4	2

$$\lambda = 1: \qquad S_A = 4 + 4 = 8, \quad S_B = 5 + 2 = 7.$$

Key message: These are candidate fits, not a claim about real data.

Penalty Strength Changes the Balance

$$\lambda = 0.1: \quad S_A = 4 + 0.1(4) = 4.4, \quad S_B = 5 + 0.1(2) = 5.2.$$

- Zero penalty: the OLS objective.
- Higher strength: more emphasis on coefficient cost.
- Excessive strength can erase useful relationships.
- Validation determines whether the restriction helps forecasting.

A Penalty Path Is Not a Stability Experiment

- **Coefficient path**: fixed sample, vary penalty strength.
- **Sample sensitivity**: fixed fitting procedure, vary fitting sample.
- Ridge usually retains nonzero weights.
- A smooth path does not prove stable forecasts across samples.

Input Units Change the Coefficient Charge

$$\beta x = (\beta/100)(100x).$$

- Rescale an input: prediction can stay identical.
- The coefficient becomes 100 times smaller.
- Squared coefficient cost becomes 10,000 times smaller.
- Absolute coefficient cost becomes 100 times smaller.

Standardization Uses the Training Sample

$$z_{ij} = \frac{x_{ij} - m_j}{s_j}.$$

- m_j: predictor j's training mean.
- $s_j > 0$: its training standard deviation, or spread.
- z_{ij}: distance from the mean in training-spread units.
- A **scaler** stores training means and spreads for later transformations.

Preprocessing Must Respect the Same Time Boundary

- **Preprocessing**: transform raw data into model inputs.
- Fit the scaler on training observations only.
- Transform validation observations with the stored scaler.
- Repeat this inside every training fold.

Key message: Future observations must not choose today's input scale.

Lasso Can Remove an Input Contribution

Lasso regression

Fit a regression with an absolute-size cost on slope weights.

- The absolute-size rule is called L_1.
- Some fitted weights can become exactly zero.
- **Sparsity**: many weights are exactly zero.
- **Feature selection**: choose which inputs remain in the prediction.

Read the Lasso Score

$$\text{lasso score} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^{p} |\beta_j|.$$

- Training-loss block: unchanged from ridge.
- Absolute values: coefficient sizes without squaring.
- Strength: how much the coefficient cost matters.
- The intercept is excluded from this penalty.

A Zero Is a Modeling Decision

- Zero weight: no contribution in this fitted prediction.
- It does not prove that the input has no economic effect.
- Related inputs can substitute for each other.
- Useful weak signals may be discarded.

Key message: A zero weight does not prove that changing this input has no effect.

Ridge and Lasso: Keep the Comparison Controlled

	Ridge	Lasso
Coefficient cost	Squared sizes	Absolute sizes
Typical result	Shrinkage	Shrinkage plus possible zeros
Question to check	Does sensitivity improve?	Is selection useful and persistent?

Key message: Use the same inputs, time boundaries and error measure.

Selection Can Flip Between Related Inputs

- Repeat the fitting procedure across nearby windows.
- Record which weights are nonzero.
- Compare predictions on common reference cases.
- Separate selection changes from material forecast changes.

Optional Extension: Elastic Net

Elastic net

Combine absolute-size and squared-size coefficient penalties.

$$\text{score} = \text{training loss} + \lambda_1 \sum_j |\beta_j| + \lambda_2 \sum_j \beta_j^2.$$

- λ_1: weight on the absolute-size cost.
- λ_2: weight on the squared-size cost.
- More tuning choices; no guarantee of better forecasts.

Keep Training, Validation and Test in Time Order

Earlier	Middle	Later
Training: fit	Validation: choose	Test: report
70% example	15% example	15% example

- Percentages are illustrative, not universal.
- Match boundaries to the horizon and information availability.
- Preserve the final test during all tuning decisions.

Tune a List of Candidate Strengths

1. Choose a sensible list, or **grid**, of strengths.
2. For each strength, fit the complete procedure on training data.
3. Predict the same validation observations; calculate RMSE.
4. Choose using validation only.

Key message: A numerical strength is meaningful only with its objective and scale.

A Validation Curve Need Not Be U-Shaped

- A **validation curve** plots strength against validation RMSE.
- The lowest measured score guides the choice.
- An edge minimum may justify a wider sensible search.
- Never inspect test RMSE to decide how to widen the grid.

Refit the Whole Procedure Before the Test

1. Keep the selected model and strength setting fixed.
2. Combine the earlier training and validation observations.
3. Refit scaling and coefficients on that enlarged fitting sample.
4. Evaluate once on the later test period.

Key message: No test observation enters scaling, fitting or selection.

Measure Sensitivity on Comparable Cases

- Use nearby training windows and the same stated fitting procedure.
- Selection frequency: fraction of refits with a nonzero weight.
- Compare predictions on common reference inputs.
- Changed conditions may need a new model, not just more shrinkage.

Write a Recommendation From Measured Evidence

- **Task**: target, horizon, origin and allowed inputs.
- **Accuracy**: errors on identical evaluation dates.
- **Sensitivity**: weights, selections and predictions across fits.
- **Decision**: state the benefit and its cost.

Key message: Do not let a model name substitute for a result.

A Small Decision Exercise

Candidate	Validation RMSE	Inputs used	Sensitivity
OLS	6.0	12	Large weight changes
Ridge	5.0	12	Smaller changes
Lasso	5.2	4	Selected names vary

Illustrative values only. Same validation dates and forecast units.

Choose before inspecting the final test.

What You Should Be Able to Explain

1. Why overlapping inputs can destabilize separate weights.
2. How training loss and coefficient penalties combine.
3. Why ridge shrinkage differs from lasso selection.
4. Why scaling and tuning must respect time order.
5. How to compare accuracy, sensitivity and simplicity.

From Restriction to an Honest Forecast Comparison

- Regularization is a candidate remedy for fitting sensitivity.
- Ridge: squared-size cost; lasso: absolute-size cost and possible zeros.
- Validate the full procedure, including preprocessing.
- Report the final test and the operating trade-offs honestly.

Key message: Useful restriction must earn its place through evidence.